Metabolomics!

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What is Metabolomics

The objective of metabolomics is to characterize in the most complete and comprehensive way the pool of small molecules which are the end product of the metabolism.

The pool of these molecules is known as metabolome.

Metabolomics aims at measuring the metabolites in a quantitative way and to characterize their relations and associations.

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2. To perform molecular phenotyping (*e.g. personalized medicine*)
3. To associate a genes to its function (*e.g. to support breeding*)
4. To study the chemical interaction in complex systems (*e.g. ecological interaction are often mediated by chemicals*)

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4. Coverage: no single analytical method is able to analyze everything.
5. Pre-analytics: the approach is so sensitive that every non-reproducible pre-analytical step will affect the data.

Section 1

Doing Metabolomics

Targeted

We use the analytical method of choice to analyze a selected set of metabolites. The list can go from around 10 to something around 500.

The measured variables are metabolites

Warning

The numbers we collect is the concentration of the individual metabolites.

Untargeted

We "throw" a raw sample in our instrument, trying to analyze as much signals as possible. How many things we are able to see is unknown.

The measured variables are **not** metabolites

Warning

The numbers we get are the peak areas of the individual features.

Why targeted, why untargeted

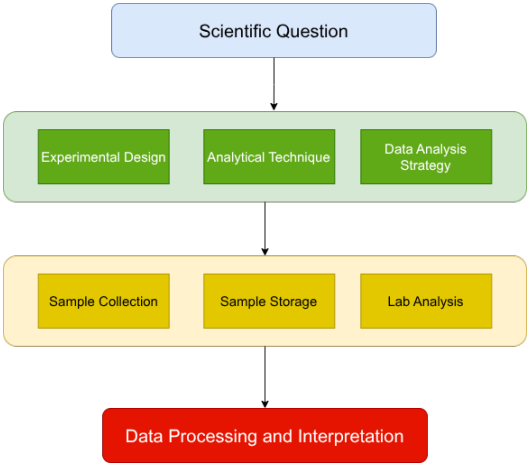
Targeted

- ✓ quantitative results are obtained "by definition"
- ✓ results can be interpreted under the light of the known metabolic pathways
- ✓ the analytical results are comparable across experiments and across laboratories (aka limited *batch effect*)
- × we basically look only to the expected

Untargeted

- ✓ holistic
- ✓ open to unexpected
- × tricky
- × hardly quantitative
- × not standardized

The ideal Metabolomics road-map



Section 2

Analytical Techniques in Metabolomics

How Metabolomics

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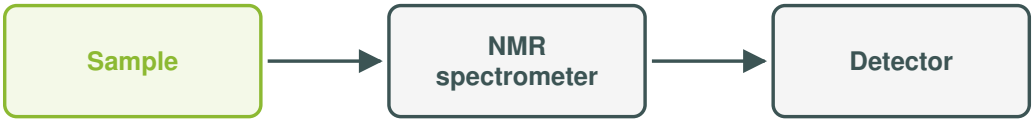
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2. **Universal** - able to see almost all classes of molecules.
3. **Sensitive** - able to see metabolites with low concentrations.
4. With an high **dynamic range** - able to measure at the same time low and high abundant compounds.

Nuclear Magnetic Resonance

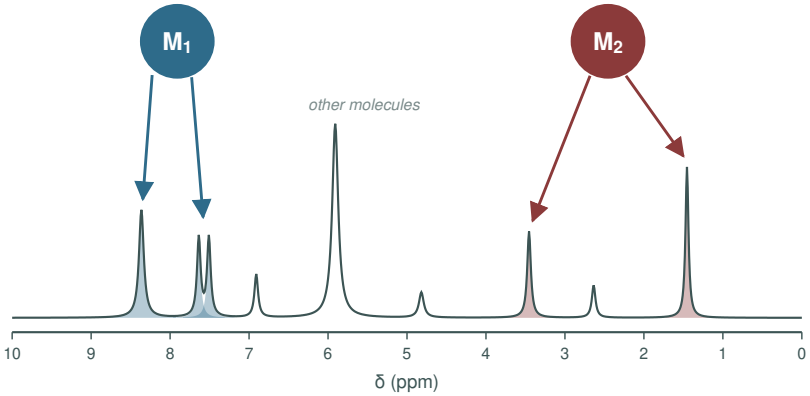


the higher the field, the better the resolution

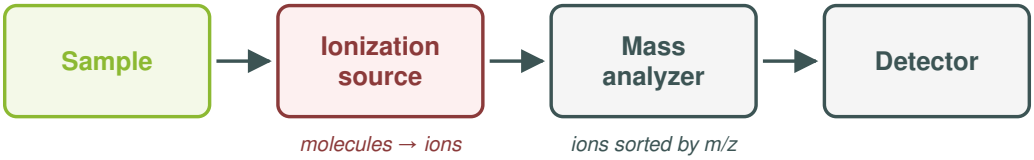
magnetic field (^1H frequency)

400 MHz
600 MHz
800 MHz
...

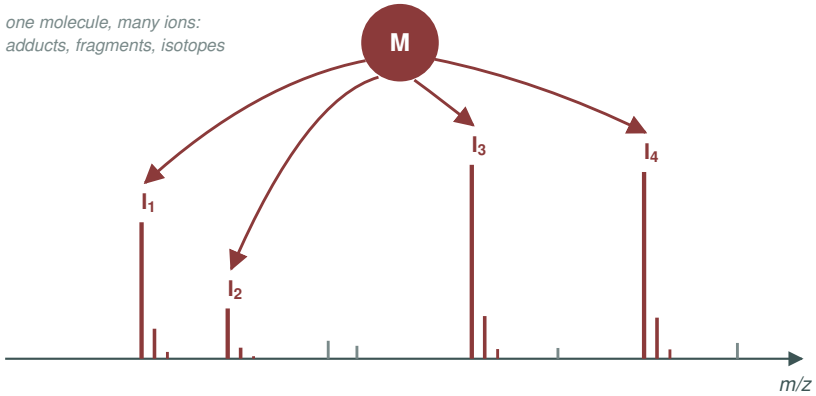
Molecules and Peaks



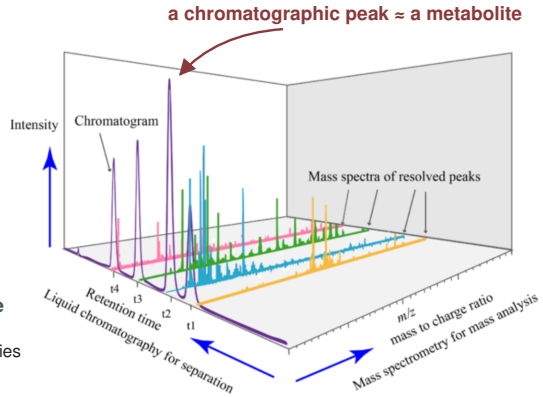
Mass Spectrometry



Molecules and Ions



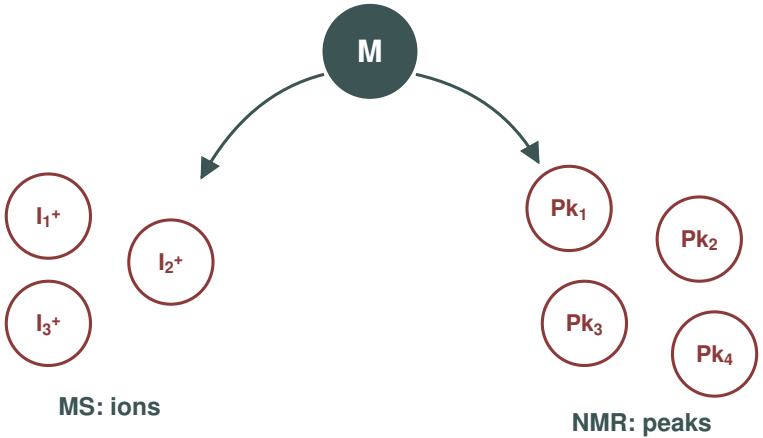
Chromatographic Separation



Retention time
depends on the chemical properties of the molecule

m/z
is the "weight" of the ions

Molecules and features



Molecules and features

In all the techniques presented so far, each metabolite give rise to a large set of different signals:

- **MS** - the different ions produced in the ionization of the metabolite.
- **NMR** - the different peaks coming from the resonance of the nuclei present in the molecule.
- **LC/GC-MS** - the different ions which are eluting at different time. These mz/rt couples are called *features*

Warning

All the variables coming from the same molecule are highly correlated.

Sources of Analytical Variability

1 - The measurement itself

Even with the perfect instrument the "numbers" will change. Try measuring your temperature 10 times in a row ;-)

2 - Chromatographic drifts

When chromatographic separation is present, elution times will slightly change from run to run. For GC the situation is better, but far from being perfect

Sources of Analytical Variability

3 - Matrix effects

In metabolomics we are loading on the instruments a complex mixture of chemicals. This complex chemical environment will affect the *response* of the molecules in an often unpredictable way. The importance of this effect is so strong in MS that this technique is almost always coupled with chromatography
With chromatography the different molecules reach the ionization source at different times, reducing the matrix effect.

4 - Sample handling

The approach is so powerful that all aspect related to sample handling/storage/extraction will affect the final result

NMR vs LC/GC MS

NMR

- ✓ Reproducible
- ✓ Inherently quantitative
- ✓ Sensitive to the chemical structure
- × Relatively insensitive
- × Dynamic range

LC/GC - MS

- ✓ Sensitive
- ✓ Universal
- ✓ With high dynamic range
- × Less reproducible (;-))

Section 3

Pitfalls and caveats

Poor planning

Aka ... I have a refrigerator of samples ... shall we give a look?

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Remember that instruments are sensitive, so errors will unavoidably show up in the final data (*garbage in, garbage out*)

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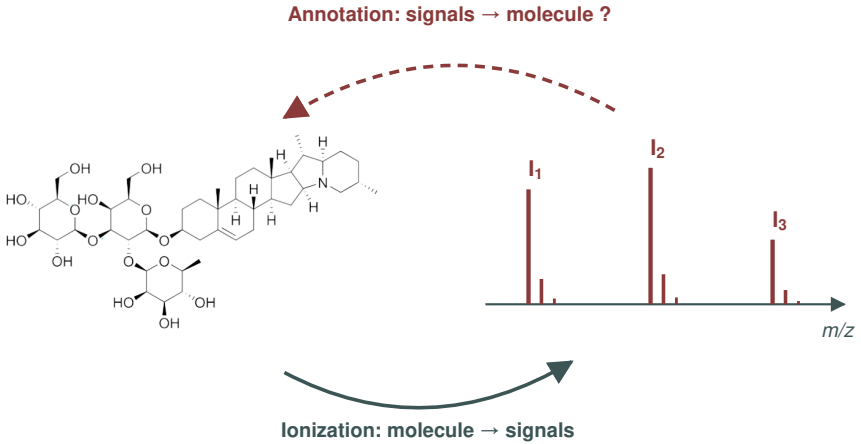
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e.g. how to sample in a representative way in GC? How to quench the metabolism of microorganisms? How to store biofluids?

Annotation ...


Annotation: signals → molecule ?
I1
I2
I3
m/z
Ionization: molecule → signals

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Chemical annotation is the current bottleneck in untargeted metabolomics

Annotation strategies

- Fragmentation experiments (in MS!)

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- ML/AI (prediction of chemical structures from spectra)

Statistical Validation

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Two-stage design: 1. hypothesis generation $\rightarrow$ 2. validation

Domain Specific Validation

Are my results fitting inside the established (domain specific) body of knowledge?

e.g. Are my markers reasonable under the light of the human metabolism?

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3. **Interdisciplinarity**
   - The analyst and/or the data analyst are your best friends ...